

Adaptive Energy-Aware Fleet Coordination for Electric Waste Collection

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Abstract—Smart waste collection uses real-time sensing to coordinate electric fleets as waste demands, traffic conditions, and vehicle states change. Vehicle assignment determines each vehicle’s workload, while routing determines its payload evolution and traffic exposure, linking them through energy consumption. This paper develops an adaptive energy-aware fleet coordination framework that represents assignment and routing within a unified graphical model. Message-passing implements dynamic-programming principles on this graph to distribute collection tasks and derive energy-efficient routes, with routing-energy metrics coordinating both decisions. The framework directly updates assignments and routes as operating conditions change, without retraining or restarting a fleet-wide search. A real-map digital-twin tested evaluates this adaptability under varying traffic, waste demand, and fleet conditions. Numerical results show **[XX% lower fleet energy consumption than BASELINE] while [COLLECTION-PERFORMANCE RESULT]**. The results further reveal a fleet-size tradeoff between per-vehicle payload and total fleet travel, providing insight into energy-efficient fleet sizing.

Index Terms—Smart waste collection, electric vehicle fleets, fleet coordination, adaptive routing, message-passing.

I. INTRODUCTION

Smart waste collection connects sensing, communication, and fleet operation to support sustainable urban services [1], [2], [3]. Internet-of-Things (IoT) infrastructure provides real-time information from geographically distributed waste bins [4], replacing fixed collection schedules with operations that respond to observed demands [5]. Recent systems use bin states and spatial information for route planning [6], coordinate multiple collection vehicles from real-time waste volumes [7], and combine collection information with vehicle operating constraints through connected vehicle systems [8]. Smart waste management has progressed from passive monitoring demand toward automated fleet operation driven by current operating conditions.

The electrification of waste collection fleets extends this development toward low-carbon urban transportation [9]. Electric vehicles are increasingly considered as alternatives to conventional refuse trucks [10], placing energy consumption at the center of fleet operation [11]. An electric refuse truck consumes energy according to its payload, speed, road conditions,

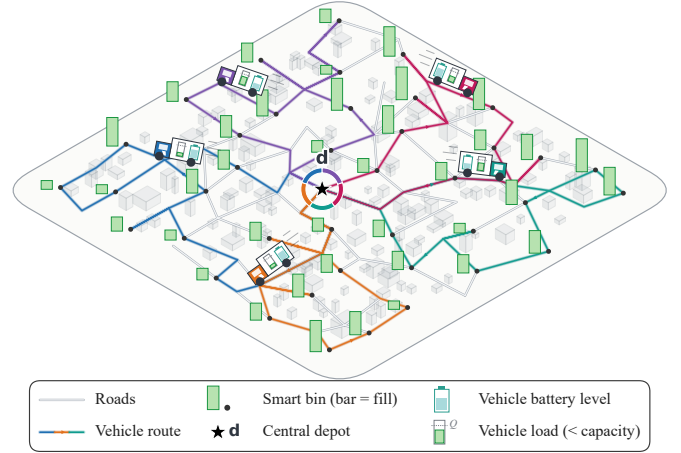


Fig. 1. Adaptive energy-aware fleet coordination for smart waste collection.

and collection operations rather than travel distance alone [12]. Vehicle assignment determines the collection workload and the waste accumulated by each vehicle, while routing determines how that payload evolves and which traffic conditions the vehicle encounters. Energy-efficient operation depends jointly on task distribution and route construction.

Fig. 1 illustrates the coupling between vehicle assignment and routing in electric waste collection. The assignment determines which bins form each vehicle’s workload, while routing determines their visiting order and the resulting payload evolution. General vehicle routing already couples customer assignment with visit ordering [13], but electric waste collection further links both decisions through load- and traffic-dependent energy consumption. The energy cost of an assignment cannot be determined until its routing problem is resolved, while the routing cost itself depends on the payload created by that assignment. Waste demand, traffic, and vehicle states also change during collection, altering this relationship after an operating plan has been constructed. Fleet coordination resolves coupled assignment and routing decisions by adapting them to changing energy costs and operating conditions.

There have been several studies that present several mechanisms for handling such combinatorial structure of fleet planning. Electric waste collection models incorporate battery and working-time constraints and use clustering and routing heuristics for larger instances [14]. Urban fleet models account for time-varying traffic and load-dependent operating costs [15]. Heuristic and metaheuristic methods construct solutions through neighborhood, insertion, and repair operations [16],

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while dynamic-programming exploits the sequential structure of routing to reduce explicit route enumeration [17]. These methods provide effective tools for assigning or routing vehicles under detailed operating constraints. However, solving assignment and routing separately prevents the assignment stage from directly accounting for the route-dependent energy cost, whereas jointly enumerating assignments and routes leads to a rapidly growing search space. The central challenge is to incorporate routing-energy information into vehicle assignment without explicitly searching over all assignment-route combinations.

Changing operating conditions make this coordination inherently dynamic. Waste collection demands, traffic, and vehicle states directly determine the energy costs used for assignment and routing, and their variation can change both decisions during the operation. Dynamic waste collection methods revise routes using real-time bin information and time-dependent travel times [18], while rolling-horizon approaches reconstruct plans as new information becomes available [19]. For electric fleets, route updates also affect the energy costs used for vehicle assignment. Changes in operating conditions affect assignment and routing through their coupled energy costs.

Machine-learning-based methods provide another approach to rapid replanning. Deep reinforcement learning shifts route construction to pretrained policies, reducing the computation performed during operation [20]. Graph neural networks similarly estimate routing costs from learned graph representations and support higher-level fleet decisions [21]. Learning has also been combined with local search to handle stochastic waste demand [22]. These approaches reduce online computation by learning parts of the routing process in advance, but their responses depend on the operating conditions represented during training. Changes in problem size and data distribution consequently degrade the generalization of neural routing models [23]. Repeated combinatorial search and pretrained models thus provide different mechanisms for replanning, but neither tracks the changing energy relationship between assignment and routing within a common computational structure.

This paper develops a distributed framework for adaptive energy-aware fleet coordination. A unified graphical model represents vehicle assignment and routing through nodes describing their decision variables, constraints, and operating states. The graph is decomposed into tractable components for distributed computation, with dynamic-programming principles characterizing the sequential decisions within each component. Message-passing framework implements these computations and exchanges information between the assignment and routing components. Instead of returning only a single optimal decision, each component produces real-valued messages that quantify the confidence of alternative decisions. In particular, the routing component extends dynamic-programming to compute decision-confidence metrics for alternative routing choices, allowing route-dependent energy information to directly influence vehicle assignment. The exchanged metrics thereby coordinate task distribution and route construction without exhaustive fleet-wide search. As operating conditions change, the same message-passing structure updates these metrics and produces new assignments and routes, providing

adaptive fleet operation through distributed computation.

The main contributions of this work are summarized as:

- **Energy-aware fleet formulation:** Electric waste collection is formulated as a coupled vehicle-assignment and routing problem that captures evolving payload, time-varying traffic, and vehicle energy consumption. A unified graphical model represents dual route-dependent energy relationship within a common framework.
- **Distributed adaptive coordination:** Dynamic-programming principles are developed for the decomposed graph, and message-passing implements the resulting computations in a distributed manner. Real-valued decision metrics exchange routing-energy information between assignment and routing, enabling both decisions to adapt to changing operating conditions. Numerical results show [XX% lower fleet energy consumption than BASELINE] while [COLLECTION-PERFORMANCE RESULT].
- **Digital-twin validation and fleet-sizing insight:** A real-map digital-twin testbed based on the Seoul road network evaluates the framework under varying traffic, waste demand, and fleet conditions. The evaluation reveals a tradeoff between reduced per-vehicle payload and increased total fleet travel as fleet size grows, identifying an energy-efficient operating region for fleet sizing.

The results establish the proposed framework as an effective solution for adaptive energy-efficient electric waste collection.

The remainder of this paper is organized as: Section II presents the system model and problem formulation. Section III presents a distributed solution to the coupled assignment and routing problem. Section IV presents the digital-twin evaluation and numerical results, and Section V concludes the paper.

II. SYSTEM MODEL

This section presents a electric waste collection model under changing waste, traffic, and vehicle conditions. The model first describes the information available at each coordination epoch, then derives load- and traffic-dependent travel energy, and finally formulates the coupled vehicle-assignment and routing task.

A. Waste collection in urban road networks

Consider an automated waste collection system with N smart bins and K electric collection vehicles. Let $\mathcal{I} = \{1, \dots, N\}$ and $\mathcal{K} = \{1, \dots, K\}$ denote the bin and vehicle index sets, respectively. Each smart bin has a known location and reports its current uncollected waste mass, and the road infrastructure provides traffic information for the urban road network [6].

Fleet coordination is updated as the operating conditions change. Let τ denote the physical time at which the current fleet plan is constructed. All information available at τ defines the current coordination epoch. During the resulting vehicle operation, $t \geq \tau$ denotes a physical time. When a vehicle traverses a road path consisting of consecutive edges, t_r denotes the physical time at which it enters the r th edge

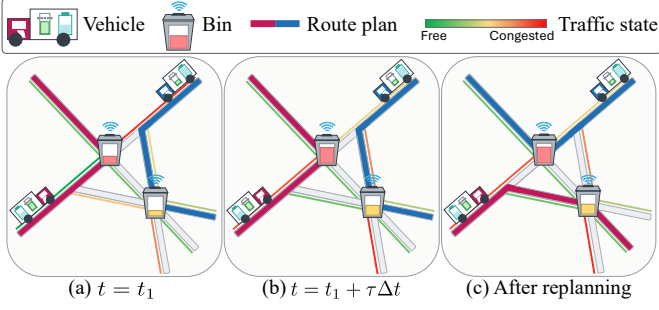


Fig. 2. Adaptive electric waste collection under time-varying traffic conditions, where updated traffic information can alter both vehicle-to-bin assignments and routes.

of that path. Thus, τ specifies the time at which the fleet problem is constructed, t describes time during the subsequent operation, and t_r tracks time along an individual road path. Let $\mathcal{I}_\tau \subseteq \mathcal{I}$ denote the bins that remain unserved at time τ . Bin $i \in \mathcal{I}_\tau$ contains uncollected waste mass $w_i(\tau)$. Each vehicle has payload capacity $\ell_{\max}$, curb mass m_0 , maximum battery energy $B_{\max}$, and minimum battery reserve $B_{\min}$. The state of vehicle $k \in \mathcal{K}$ at physical time t is

$$\mathbf{y}_k(t) = [p_k(t) \quad \ell_k(t) \quad b_k(t)]^\top, \quad (1)$$

where $p_k(t)$, $\ell_k(t)$, and $b_k(t)$ denote its position, payload, and remaining battery energy, respectively. At the current coordination time τ , the available payload capacity of vehicle k is $\ell_{\max} - \ell_k(\tau)$. Initially, all vehicles depart from a common depot d with zero payload and battery energy $B_{\max}$. The fleet plan at τ is constructed from the remaining waste demands $\{w_i(\tau)\}_{i \in \mathcal{I}_\tau}$, the vehicle states $\{\mathbf{y}_k(\tau)\}_{k \in \mathcal{K}}$, and the traffic information available at that time. A subsequent change in these quantities defines a new coordination time and a new fleet-planning instance.

Fig. 2 shows how the operating state connects vehicle assignment and routing. Assignment determines the collection workload of each vehicle and the payload accumulated during collection. Routing determines the order in which this payload evolves and the traffic conditions encountered along the trip. Waste demand, traffic, and vehicle states therefore determine how vehicle assignment and routing jointly affect fleet energy consumption.

B. Load- and traffic-dependent energy model

The energy cost of vehicle travel depends on both the carried payload and the traffic encountered along the road network. The depot and bin locations are embedded in a directed road graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each edge $e \in \mathcal{E}$ represents a directed road segment with length L_e and inclination angle θ_e . The fleet plan is constructed at the current coordination time τ . For travel occurring after τ , let t denote the physical departure time of a vehicle from a location. When the corresponding road path consists of consecutive edges $(e_1, \dots, e_M)$, let t_r denote the time at which the vehicle enters edge e_r . Thus, $t_1 = t \geq \tau$, and the subsequent edge-entry times are determined by the preceding edge traversal times.

Based on the traffic information available at τ , let $T_e(t)$ denote the estimated traversal time of edge e for a vehicle entering it at time t . The corresponding average speed is

$$v_e(t) = \frac{L_e}{T_e(t)}. \quad (2)$$

The electric refuse-truck model in [12] relates vehicle mass, road geometry, and traffic conditions to battery consumption. A vehicle entering edge e at time t with payload ℓ has total mass $m_0 + \ell$. Its longitudinal tractive force is

$$F_e(t, \ell) = (m_0 + \ell)g(C_{rr} \cos \theta_e + \sin \theta_e) + \frac{1}{2}\rho C_d A_f v_e(t)^2, \quad (3)$$

where g is the gravitational acceleration, C_{rr} is the rolling-resistance coefficient, ρ is the air density, C_d is the aerodynamic drag coefficient, and A_f is the frontal area. The drivetrain converts this mechanical power differently under traction and regenerative braking. Define the corresponding power-conversion factor as

$$\gamma(F) = \begin{cases} 1/\eta_{\text{drv}}, & F \geq 0, \\ \eta_{\text{reg}}, & F < 0, \end{cases} \quad (4)$$

where η_{drv} and η_{reg} denote the drivetrain and regenerative efficiencies, respectively. The battery power associated with vehicle motion over edge e is given by

$$P_e(t, \ell) = \frac{\gamma(F_e(t, \ell))F_e(t, \ell)v_e(t)}{1000}, \quad (5)$$

where the division by 1000 converts power from watts to kilowatts. A positive $P_e(t, \ell)$ represents battery consumption for traction, whereas a negative value represents energy recovery through regenerative braking. Including auxiliary power P_{aux} , average battery energy consumed over edge e is expressed as

$$E_e^{\text{avg}}(t, \ell) = \frac{T_e(t)}{3600} [P_e(t, \ell) + P_{\text{aux}}], \quad (6)$$

where the division by 3600 converts the traversal time from seconds to hours, so that the energy is measured in kWh. The resulting edge energy depends on payload through the vehicle mass and on traffic through both the traversal speed and traversal time. In addition to the energy associated with average edge traversal, congested traffic can induce repeated braking and re-acceleration, particularly near road intersections. To capture this consumption and add it into the battery energy, let T_e^{ref} denote the nominal traversal time of edge e and let ΔT_e specify a tolerance around this nominal condition. We define the normalized traversal-time deviation as

$$z_e(t) = \frac{T_e(t) - T_e^{\text{ref}}}{\Delta T_e}, \quad (7)$$

and introduce the congestion penalty $\phi_e(t) = [z_e(t) - 1]^+$, where $[x]^+ = \max\{x, 0\}$. Hence, normal variations in traversal time do not incur an additional energy cost, whereas sufficiently prolonged one increases the penalty linearly. The resulting edge energy is augmented as

$$E_e(t, \ell) = E_e^{\text{avg}}(t, \ell) + \kappa_{\text{sg}}(m_0 + \ell)\phi_e(t), \quad (8)$$

where κ_{sg} is the stop-and-go energy coefficient. The additional term represents the increased energy associated with braking and subsequent acceleration under congested traffic and increases with vehicle mass, reflecting the greater energy required to repeatedly accelerate a more heavily loaded vehicle. Thus, traffic affects the edge energy not only through the average traversal speed and duration, but also through the additional stop-and-go operation induced by severe congestion.

The edge-level energy model is extended to travel between service locations. Let a and b denote two bins or the depot, and let Π_{ab} denote the set of feasible directed road paths from a to b . Consider a path $\pi = (e_1, \dots, e_M) \in \Pi_{ab}$ for a vehicle departing from a at time t with payload ℓ . As defined above, t_r denotes the entry time to edge e_r , with $t_1 = t$. The subsequent edge-entry times follow

$$t_r = t_{r-1} + T_{e_{r-1}}(t_{r-1}), \quad r = 2, \dots, M. \quad (9)$$

Since no waste is collected between two service locations, the payload remains ℓ along the path. The energy for path π is

$$E_\pi(t, \ell) = \sum_{r=1}^M E_{e_r}(t_r, \ell). \quad (10)$$

Also, the minimum travel energy from a to b is

$$E_{ab}(t, \ell) = \min_{\pi \in \Pi_{ab}} E_\pi(t, \ell), \quad (11)$$

and $T_{ab}(t, \ell)$ denotes the travel time along the corresponding energy-minimizing path. These two quantities describe the energy and time required to move between service locations for a given departure time and payload, and are used to construct the collection routes.

C. Vehicle assignment and energy-aware routing

At replanning epoch τ , the remaining collection tasks are distributed among available vehicles, and a visiting order is determined for the bins assigned to each vehicle. Let $\mathcal{I}_{k,\tau} \subseteq \mathcal{I}_\tau$ denote the bins assigned to vehicle k . Every remaining bin is assigned to exactly one vehicle, and the resulting workload cannot exceed its available payload capacity. Hence, the corresponding constraints are given as

$$\bigcup_{k \in \mathcal{K}} \mathcal{I}_{k,\tau} = \mathcal{I}_\tau, \quad (12a)$$

$$\mathcal{I}_{k,\tau} \cap \mathcal{I}_{k',\tau} = \emptyset, \quad \forall k \neq k', \quad (12b)$$

$$\ell_k(t_\tau) + \sum_{i \in \mathcal{I}_{k,\tau}} w_i(t_\tau) \leq \ell_{\max}, \quad \forall k \in \mathcal{K}. \quad (12c)$$

Let $n_{k,\tau} = |\mathcal{I}_{k,\tau}|$ denote the number of bins assigned to vehicle k . Its remaining route consists of these bins followed by a return to the depot, resulting in $H_{k,\tau} = n_{k,\tau} + 1$ travel transitions. The route is represented by

$$\mathbf{a}_{k,\tau} = (a_{k,0}, a_{k,1}, \dots, a_{k,H_{k,\tau}}). \quad (13)$$

where $a_{k,0} = p_k(t_\tau)$ is the current vehicle position, $a_{k,1}, \dots, a_{k,n_{k,\tau}}$ contain every bin in $\mathcal{I}_{k,\tau}$ exactly once, and $a_{k,H_{k,\tau}} = d$ represents the return to the depot. The energy of each route transition depends on the departure time and the payload carried over that transition. Let $t_{k,h}$, $\ell_{k,h}$, and

$b_{k,h}$ denote the departure time, payload, and remaining battery energy, respectively, after vehicle k services location $a_{k,h}$. The route is initialized from the vehicle state observed at t_τ , with $t_{k,0} = t_\tau$, $\ell_{k,0} = \ell_k(t_\tau)$, and $b_{k,0} = b_k(t_\tau)$. For transition h , the departure time $t_{k,h-1}$ and payload $\ell_{k,h-1}$ are used as t and ℓ , respectively, in $E_{ab}(t, \ell)$ and $T_{ab}(t, \ell)$. Servicing bin i takes s_i seconds and consumes lifting and compaction energy e_i^{lc} . The stationary auxiliary systems additionally consume power P_{svc} during service. The resulting service energy is

$$E_i^{\text{svc}} = e_i^{\text{lc}} + P_{\text{svc}} \frac{s_i}{3600}, \quad (14)$$

where division by 3600 converts the service time from seconds to hours. For the depot, $w_d(t_\tau) = 0$, $s_d = 0$, and $E_d^{\text{svc}} = 0$.

For transition h , vehicle k departs from $a_{k,h-1}$ at time $t_{k,h-1}$ with payload $\ell_{k,h-1}$, travels to $a_{k,h}$, and then services the destination. The departure time after service is updated as

$$t_{k,h} = t_{k,h-1} + T_{a_{k,h-1}, a_{k,h}}(t_{k,h-1}, \ell_{k,h-1}) + s_{a_{k,h}}. \quad (15)$$

The collected waste is then added to the vehicle payload according to

$$\ell_{k,h} = \ell_{k,h-1} + w_{a_{k,h}}(t_\tau), \quad (16)$$

and the battery state for $h = 1, \dots, H_{k,\tau}$ becomes

$$b_{k,h} = b_{k,h-1} - E_{a_{k,h-1}, a_{k,h}}(t_{k,h-1}, \ell_{k,h-1}) - E_{a_{k,h}}^{\text{svc}}. \quad (17)$$

The travel energy of transition h is evaluated using the payload $\ell_{k,h-1}$ carried before collecting waste at $a_{k,h}$. The newly collected waste therefore affects the energy consumption from the next transition onward. Since the payload increases monotonically along the collection route, the aggregate capacity constraint in (12) guarantees the payload limit throughout the route. The total energy consumed by vehicle k over its remaining route is given by

$$E_k(t_\tau, \mathbf{a}_{k,\tau}) = \sum_{h=1}^{H_{k,\tau}} E_{a_{k,h-1}, a_{k,h}}(t_{k,h-1}, \ell_{k,h-1}) + \sum_{h=1}^{H_{k,\tau}} E_{a_{k,h}}^{\text{svc}}. \quad (18)$$

The fleet-coordination problem at replanning epoch τ is formulated as

$$\begin{aligned} \text{(P1)} \quad & \min_{\{\mathcal{I}_{k,\tau}\}, \{\mathbf{a}_{k,\tau}\}} \sum_{k \in \mathcal{K}} E_k(t_\tau, \mathbf{a}_{k,\tau}) \\ & \text{subject to} \quad (12), (15), (16), (17), \\ & \quad B_{\min} \leq b_{k,h} \leq B_{\max}, \quad \forall k, h. \end{aligned} \quad (19)$$

Problem (P1) couples vehicle assignment and routing through energy consumption. The assignment determines the bins collected by each vehicle and thereby its payload evolution. The visiting order determines when each payload is carried and which traffic conditions are encountered. Hence, the energy cost of an assignment is determined only after its route is evaluated, while these route-dependent energy costs determine which assignments are energy efficient. At $\tau = 0$, (P1) determines the initial fleet plan from the observed waste, traffic, and vehicle states. At each subsequent replanning

epoch, updated system information changes the travel costs, the remaining collection tasks, and vehicle states in the same formulation. The fleet plan can adapt to changing operating conditions without changing the underlying structure.

Directly solving **(P1)** is challenging since the energy cost of each assignment depends on its corresponding route, while route construction depends on the assigned bins and their resulting payload evolution. As a result, evaluating candidate assignments entails the corresponding routing problems. Also, its states further depend on the sequential evolution of payload, time, and battery energy. The resulting joint solution space grows rapidly with both the number of remaining bins and the fleet size. The following section develops a distributed solution that coordinates vehicle assignment and state-dependent routing without enumerating their fleet-wide combinations.

III. PROPOSED ALGORITHM

The structure of **(P1)** provides a natural decomposition between fleet-level assignment and vehicle-level routing. Once a set of bins is assigned to a vehicle, its route evolves sequentially through the assigned bins, and its transition cost depends only on the current route state and the next visit. This sequential structure admits a dynamic-programming (DP) representation for each vehicle. A fleet-wide DP, however, would have to track the assignments and partial routes of all vehicles simultaneously, resulting in a state space that grows rapidly with N_τ and K .

The proposed approach therefore decomposes the fleet-wide problem into an assignment graph and vehicle-wise routing trellises. The assignment graph enforces exclusive bin assignment and vehicle capacity, while each trellis applies the DP principle to evaluate the route consequences of the bins associated with one vehicle. Importantly, the routing computation returns real-valued metrics for candidate assignment decisions rather than only the minimum-energy route. These metrics summarize how each candidate assignment affects the vehicle-level route cost and thereby provide the information needed to coordinate the otherwise separated assignment and routing computations.

The resulting graphical model is solved through distributed message passing [24]. Messages propagate the decision metrics between the assignment graph and vehicle trellises, so that each component performs its local computation while accounting for the energy consequences obtained from the other component. When traffic, waste demand, or vehicle states change at a new replanning epoch, the corresponding local quantities are updated and the same decomposition is applied to the new fleet state.

For compactness, let $N_\tau = |\mathcal{I}_\tau|$, $w_i \equiv w_i(t_\tau)$, and $q_k = \ell_{\max} - \ell_k(t_\tau)$. The epoch index is suppressed from the route quantities throughout this section.

(P1) couples vehicle assignment with state-dependent routing. Each unserved bin must be assigned to one capacity-feasible vehicle, while its energy consequence depends on the visitation order, evolving payload, and traffic conditions. A fleet-wide dynamic program must jointly represent every assignment and all partial vehicle routes, causing its state space to grow rapidly with N_τ and K .

This structure motivates a vehicle-wise factor graph decomposition coordinated through message-passing [24]. Assignment and capacity conditions remain local across replanning epochs, while traffic-dependent route energy is evaluated by an independent trellis for each vehicle. The binary assignment graph and vehicle trellises cooperatively exchange their local results, allowing the remaining assignments and routes to be updated using the latest traffic and vehicle states.

For compactness, let $N_\tau = |\mathcal{U}_\tau|$, $w_i \equiv w_i(t_\tau)$, $q_k = Q - \ell_k(t_\tau)$, and suppress the epoch index on route quantities, so that $\mathbf{r}_k \equiv \mathbf{r}_{k,\tau}$ and $H_k \equiv H_{k,\tau}$ throughout this section.

A. Graph model decomposition

For every $i \in \mathcal{U}_\tau$ and $k \in \mathcal{K}$, define the variable $b_{ik} \in \{0, 1\}$ which equals one when bin i is assigned to vehicle k . Stacking these variables gives the assignment matrix $\mathbf{B} = [b_{ik}]$.

Exclusive assignment of bin i is represented by

$$I_i(\mathbf{b}_{i,:}) = \begin{cases} 0, & \sum_{k \in \mathcal{K}} b_{ik} = 1 \\ -\infty, & \text{otherwise} \end{cases} \quad (20)$$

and the capacity condition of vehicle k is represented by

$$V_k(\mathbf{b}_{:,k}) = \begin{cases} 0, & \sum_{i \in \mathcal{U}_\tau} w_i b_{ik} \leq q_k, \sum_{i \in \mathcal{U}_\tau} b_{ik} \geq 1, \\ -\infty, & \text{otherwise.} \end{cases} \quad (21)$$

Accordingly, I_i coordinates the vehicles competing for bin i , while V_k coordinates the bins sharing the residual capacity of vehicle k .

For each vehicle k , routing factor R_k connects the assignment variables in $\mathbf{b}_{:,k}$ to the terminal state of its vehicle-specific trellis. The factor returns a finite value only when the terminal state has visited exactly the selected bins and returned to the depot. Starting from the measured state $\mathbf{y}_k(t_\tau)$, the trellis evaluates feasible visitation orders using the traffic- and load-dependent energy costs defined in the system model.

Fig. 3 presents the resulting factor graph. Fig. 3a shows the assignment graph and one of the K vehicle-specific trellises, Fig. 3b shows the messages among factors, and Fig. 3c shows the message exchanges during the replanning operation.

B. Algorithm derivations

A message $\mu_{A \rightarrow B}(b_{ik} = c)$ represents the preference sent from node A to node B when $b_{ik} = c$. Since b_{ik} is binary, each message connected to b_{ik} is represented by the difference between its values at one and zero [25]. The messages exchanged within the binary assignment graph are

$$\begin{aligned} \eta_{ik} &= \mu_{I_i \rightarrow b_{ik}}(1) - \mu_{I_i \rightarrow b_{ik}}(0) \\ \omega_{ik} &= \mu_{b_{ik} \rightarrow I_i}(1) - \mu_{b_{ik} \rightarrow I_i}(0) \\ \phi_{ik} &= \mu_{V_k \rightarrow b_{ik}}(1) - \mu_{V_k \rightarrow b_{ik}}(0) \\ \gamma_{ik} &= \mu_{b_{ik} \rightarrow V_k}(1) - \mu_{b_{ik} \rightarrow V_k}(0) \\ \rho_{ik} &= \mu_{b_{ik} \rightarrow R_k}(1) - \mu_{b_{ik} \rightarrow R_k}(0) \\ \delta_{ik} &= \mu_{R_k \rightarrow b_{ik}}(1) - \mu_{R_k \rightarrow b_{ik}}(0). \end{aligned} \quad (22)$$

At variable node b_{ik} , each outgoing message is obtained by adding the messages received from the other two factors and thus $\omega_{ik} = \phi_{ik} + \delta_{ik}$, $\gamma_{ik} = \eta_{ik} + \delta_{ik}$, and $\rho_{ik} = \eta_{ik} + \phi_{ik}$.

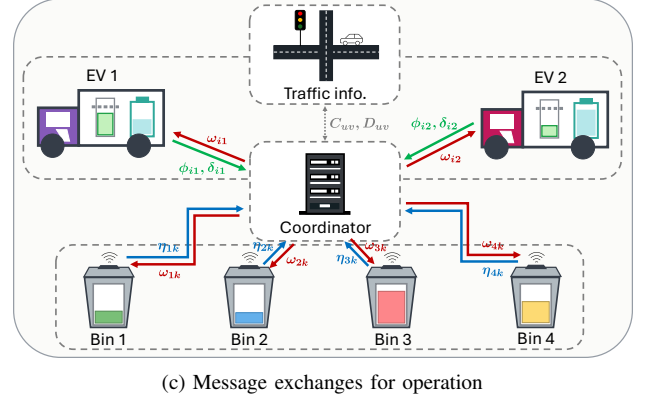
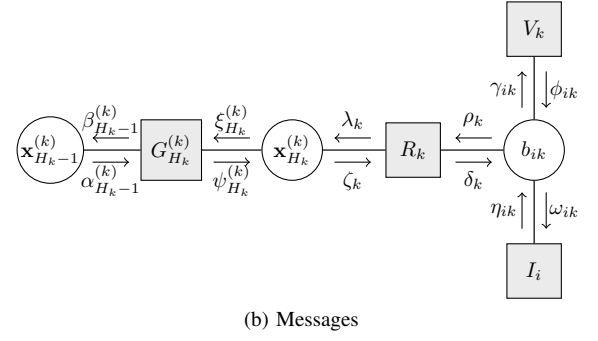
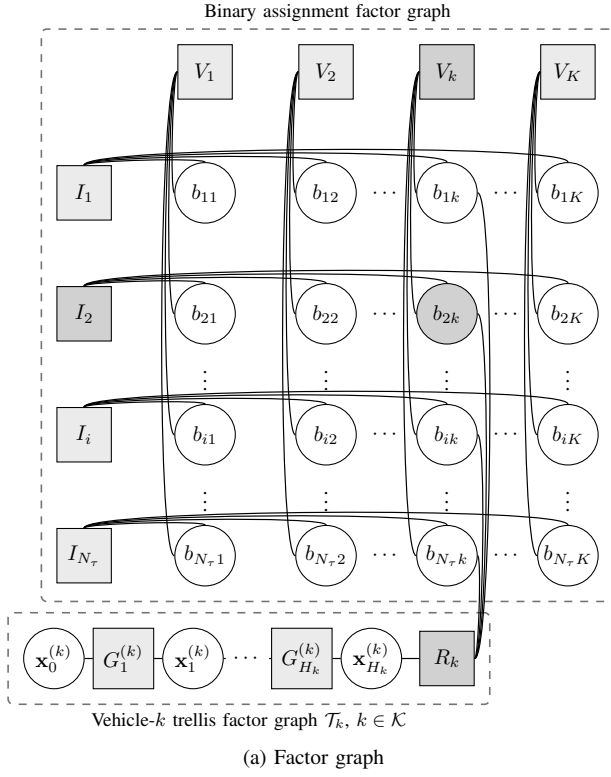


Fig. 3. Factor graph of the cooperative assignment and routing structure. For each vehicle k , the routing-consistency factor R_k connects its assignment variables to the terminal state of the corresponding vehicle-specific trellis.

For vehicle k , let $\mathbf{m}_h^{(k)} \in \{0, 1\}^{N_r}$ denote the visited-bin record after h route transitions. The trellis state is defined as

$$\mathbf{x}_h^{(k)} = \left[(\mathbf{m}_h^{(k)})^\top \quad t_{k,h} \quad \mathbf{y}_{k,h}^\top \right]^\top, \quad h = 0, \dots, H_k, \quad (23)$$

where $\mathbf{y}_{k,h}$ is the route-propagated vehicle state in (??). Hence, $\mathbf{x}_h^{(k)}$ augments the physical state with the visited-bin record and departure time required for route optimization. Its initial condition is

$$\mathbf{m}_0^{(k)} = \mathbf{0}, \quad t_{k,0} = t_\tau, \quad \mathbf{y}_{k,0} = \mathbf{y}_k(t_\tau). \quad (24)$$

At a feasible terminal state, $\mathbf{m}_{H_k}^{(k)} = \mathbf{b}_{:,k}$ and the location component of $\mathbf{y}_{k,H_k}$ equals the depot d .

Accordingly, the routing-consistency factor is defined as

$$R_k(\mathbf{b}_{:,k}, \mathbf{x}_{H_k}^{(k)}) = \begin{cases} 0, & \mathbf{m}_{H_k}^{(k)} = \mathbf{b}_{:,k}, a_{k,H_k} = d \\ -\infty, & \text{otherwise.} \end{cases} \quad (25)$$

Thus, R_k is a hard consistency factor and does not itself contain the route-energy cost. The route energy is accumulated through the transition factors $G_h^{(k)}$.

Unlike the binary assignment variables, $\mathbf{x}_h^{(k)}$ takes values over the feasible combinations of visited bins, location, time, payload, and battery energy. Messages connected to a trellis state, $\lambda_k(\mathbf{x}_{H_k}^{(k)})$ and $\zeta_k(\mathbf{x}_{H_k}^{(k)})$ between R_k and $\mathbf{x}_{H_k}^{(k)}$, together with $\alpha_h^{(k)}(\mathbf{x}_h^{(k)})$, $\beta_h^{(k)}(\mathbf{x}_h^{(k)})$, $\xi_h^{(k)}(\mathbf{x}_h^{(k)})$, and $\psi_h^{(k)}(\mathbf{x}_h^{(k)})$ within the trellis, are therefore functions over feasible state values rather than differences between binary values.

The assignment factor I_i receives ω_{ih} from the variables

associated with competing vehicles. Its outgoing message is

$$\eta_{ik} = -\max_{h \in \mathcal{K} \setminus \{k\}} \omega_{ih}. \quad (26)$$

The capacity factor V_k receives γ_{jk} from the bin variables associated with vehicle k . Its outgoing message is

$$\phi_{ik} = \begin{cases} P_k^{-i}(q_k - w_i) - P_{k,+}^{-i}(q_k), & w_i \leq q_k, \\ -\infty, & w_i > q_k, \end{cases} \quad (27)$$

Here, $P_k^{-i}(L)$ returns the largest sum of the incoming messages γ_{jk} that can be selected from bins $j \neq i$ within capacity L , where the empty selection is allowed. In contrast, $P_{k,+}^{-i}(L)$ is evaluated over nonempty selections:

$$P_{k,+}^{-i}(L) = \max_{\substack{\emptyset \neq S \subseteq \mathcal{U}_r \setminus \{i\} \\ \sum_{j \in S} w_j \leq L}} \sum_{j \in S} \gamma_{jk}. \quad (28)$$

If no nonempty feasible selection exists, then $P_{k,+}^{-i}(L) = -\infty$. Appendix A presents its Pareto-frontier evaluation, while Appendix B derives (26)-(27).

At cooperative round r , the routing messages returned by the vehicle trellises form the assignment surrogate

$$(\mathbf{P2})^{[r]} \max_{\mathbf{B}} \sum_{i \in \mathcal{U}_r} I_i(\mathbf{b}_{i,:}) + \sum_{k \in \mathcal{K}} V_k(\mathbf{b}_{:,k}) + \sum_{i \in \mathcal{U}_r} \sum_{k \in \mathcal{K}} \delta_{ik}^{[r-1]} b_{ik}. \quad (29)$$

Hard factors retain unique assignment, vehicle capacity, and nonempty-vehicle conditions of (P1), while $\delta_{ik}^{[r-1]}$ conveys the route-energy effect returned by vehicle k . The resulting

preference for assigning bin i to vehicle k is $\eta_{ik} + \phi_{ik} + \delta_{ik}$.

A feasible transition from $\mathbf{x}_{h-1}^{(k)}$ to $\mathbf{x}_h^{(k)}$ follows the physical state propagation in (??): it advances the departure time, updates $\mathbf{y}_{k,h}$ through the new location, payload, and battery energy, and marks the serviced bin in $\mathbf{m}_h^{(k)}$.

Transition factor $G_h^{(k)}$ connects $\mathbf{x}_{h-1}^{(k)}$ and $\mathbf{x}_h^{(k)}$. If the transition travels from location u to location v at departure time t with payload ℓ , its value is

$$G_h^{(k)}(\mathbf{x}_{h-1}^{(k)}, \mathbf{x}_h^{(k)}) = \begin{cases} -C_{uv}(t, \ell) - C_v^{\text{svc}}, & \text{feasible} \\ -\infty, & \text{otherwise.} \end{cases} \quad (30)$$

Thus, every feasible branch contributes the negative of the travel and service energy introduced in (11) and (14). For a complete feasible route,

$$\sum_{h=1}^{H_k} G_h^{(k)}(\mathbf{x}_{h-1}^{(k)}, \mathbf{x}_h^{(k)}) = -E_k(t_\tau, \mathbf{r}_k). \quad (31)$$

Maximizing the accumulated trellis metric therefore minimizes the vehicle energy in (18).

The forward recursion from $G_h^{(k)}$ to $\mathbf{x}_h^{(k)}$ is

$$\psi_h^{(k)}(\mathbf{x}_h^{(k)}) = \max_{\mathbf{x}_{h-1}^{(k)}} [G_h^{(k)}(\mathbf{x}_{h-1}^{(k)}, \mathbf{x}_h^{(k)}) + \alpha_{h-1}^{(k)}(\mathbf{x}_{h-1}^{(k)})] \quad (32)$$

and each internal state forwards this result to the next transition factor

$$\alpha_h^{(k)}(\mathbf{x}_h^{(k)}) = \psi_h^{(k)}(\mathbf{x}_h^{(k)}), \quad h = 1, \dots, H_k - 1. \quad (33)$$

The terminal state sends the complete forward metric to R_k

$$\zeta_k(\mathbf{x}_{H_k}^{(k)}) = \psi_{H_k}^{(k)}(\mathbf{x}_{H_k}^{(k)}). \quad (34)$$

Routing factor R_k combines the terminal route metric with the incoming assignment messages ρ_{ik} . Its message to the terminal state is

$$\lambda_k(\mathbf{x}_{H_k}^{(k)}) = \max_{\mathbf{b}_{:,k}} [R_k(\mathbf{b}_{:,k}, \mathbf{x}_{H_k}^{(k)}) + \sum_{i \in \mathcal{U}_\tau} \rho_{ik} b_{ik}]. \quad (35)$$

Since R_k is finite only when the terminal visited-bin record equals $\mathbf{b}_{:,k}$ and the vehicle has returned to the depot, λ_k introduces the binary assignment preferences into the trellis.

The backward recursion begins with

$$\xi_{H_k}^{(k)}(\mathbf{x}_{H_k}^{(k)}) = \lambda_k(\mathbf{x}_{H_k}^{(k)}) \quad (36)$$

and proceeds according to

$$\beta_{h-1}^{(k)}(\mathbf{x}_{h-1}^{(k)}) = \max_{\mathbf{x}_h^{(k)}} [G_h^{(k)}(\mathbf{x}_{h-1}^{(k)}, \mathbf{x}_h^{(k)}) + \xi_h^{(k)}(\mathbf{x}_h^{(k)})] \quad (37)$$

with

$$\xi_h^{(k)}(\mathbf{x}_h^{(k)}) = \beta_h^{(k)}(\mathbf{x}_h^{(k)}), \quad h = 1, \dots, H_k - 1. \quad (38)$$

For $b \in \{0, 1\}$, the route factor evaluates the best complete trellis configuration satisfying $b_{ik} = b$

$$\begin{aligned} \mu_{R_k \rightarrow b_{ik}}(b) &= \max_{\mathbf{b}_{:,k} \setminus b_{ik}, \mathbf{x}_{H_k}^{(k)}} [R_k(\mathbf{b}_{:,k}, \mathbf{x}_{H_k}^{(k)}) + \zeta_k(\mathbf{x}_{H_k}^{(k)}) + \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} \rho_{jk} b_{jk}]. \end{aligned} \quad (39)$$

Algorithm 1 Cooperative Assignment and Routing

Require: Current states, traffic costs, $\epsilon, \varepsilon, R_{\max}$

Ensure: $\mathbf{B}, \{\mathbf{r}_k\}$

```

1: Initialize feasible  $\mathbf{B}^{[0]}$  and routes
2: Initialize  $\eta_{ik}, \phi_{ik},$  and  $\delta_{ik}$ 
3: for  $r = 1, \dots, R_{\max}$  do
4:   Update  $\omega_{ik}$  and  $\gamma_{ik}$ 
5:   Update and damp  $\eta_{ik}$  and  $\phi_{ik}$ 
6:   Update  $\rho_{ik}$ 
7:   for all  $k \in \mathcal{K}$  in parallel do
8:     Update  $G_h^{(k)}$  using  $C_{uv}$  and  $D_{uv}$ 
9:     Compute  $\alpha, \psi, \zeta, \lambda, \xi,$  and  $\beta$ 
10:    Compute and damp  $\delta_{ik}$  by SOVA
11:   end for
12:   Decode feasible  $\mathbf{B}_c^{[r]}$ 
13:   Reconstruct candidate routes
14:   if feasible and (44) holds then
15:     Accept the candidate
16:   else
17:     Retain the previous solution
18:   end if
19:   if (45) holds then
20:     break
21:   end if
22: end for
23: return  $\mathbf{B}^{[r]}, \{\mathbf{r}_k^{[r]}\}$ 

```

The SOVA [26] output is the difference between the two conditioned route metrics

$$\delta_{ik} = \mu_{R_k \rightarrow b_{ik}}(1) - \mu_{R_k \rightarrow b_{ik}}(0). \quad (40)$$

A positive δ_{ik} indicates that the combined assignment and energy metric favors including bin i in the route of vehicle k , while a negative value favors excluding it. Appendix C gives the derivation of the messages by the trellis.

C. Convergent cooperative updates

The binary factor graph contains cycles because each b_{ik} connects the assignment, capacity, and routing factors. Damping is applied to the three factor-to-variable binary messages to suppress oscillation. For a feasible capacity update,

$$\eta_{ik}^{[r]} = (1 - \epsilon) \eta_{ik}^{[r-1]} - \epsilon \max_{h \in \mathcal{K} \setminus \{k\}} \omega_{ih}^{[r]} \quad (41)$$

$$\phi_{ik}^{[r]} = (1 - \epsilon) \phi_{ik}^{[r-1]} + \epsilon [P_k^{-i}(q_k - w_i) - P_k^{-i}(q_k)] \quad (42)$$

$$\delta_{ik}^{[r]} = (1 - \epsilon) \delta_{ik}^{[r-1]} + \epsilon [\mu_{R_k \rightarrow b_{ik}}^{[r]}(1) - \mu_{R_k \rightarrow b_{ik}}^{[r]}(0)]. \quad (43)$$

Here, $\epsilon \in (0, 1]$ is the damping factor. When $w_i > q_k$, $\phi_{ik}^{[r]}$ is set directly to $-\infty$. Messages $\omega_{ik}, \gamma_{ik},$ and ρ_{ik} are then recomputed from $\phi_{ik} + \delta_{ik}$ and $\eta_{ik} + \phi_{ik}$.

The candidate assignment is decoded from $\eta_{ik}^{[r]} + \phi_{ik}^{[r]} + \delta_{ik}^{[r]}$ subject to (20) and (21). Complete candidate routes are reconstructed under the current traffic and vehicle states. A candidate is accepted only when every route is feasible and

$$\sum_{k \in \mathcal{K}} E_k(t_\tau, \mathbf{r}_{k,c}^{[r]}) < \sum_{k \in \mathcal{K}} E_k(t_\tau, \mathbf{r}_k^{[r-1]}). \quad (44)$$

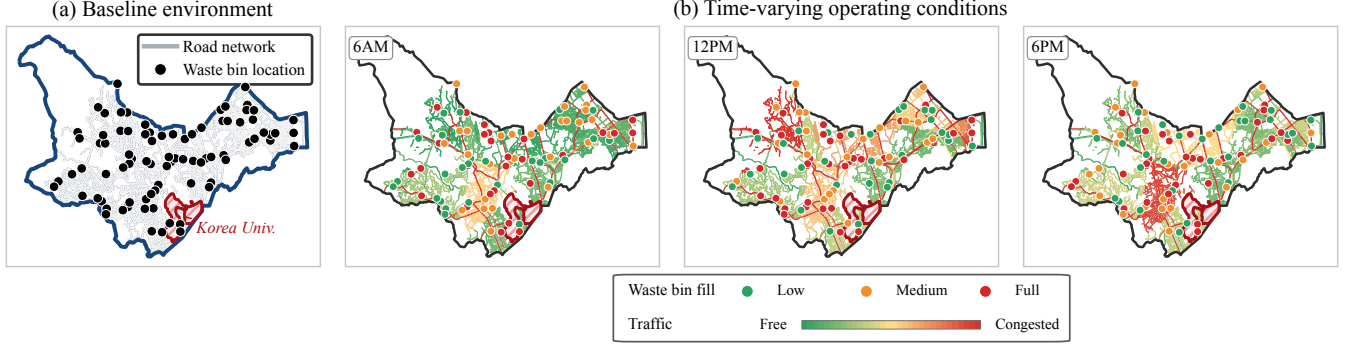


Fig. 4. Simulation environment for performance evaluation. (a) Baseline environment of the waste collection system in Seongbuk-gu, Seoul. (b) Visualization of time-varying operating conditions based on Seoul Transport Operation and Information Service (TOPIS) [27] with randomly distributed waste load.

The iterations terminate when the assignment remains unchanged and

$$\max_{i,k} \{ |\eta_{ik}^{[r]} - \eta_{ik}^{[r-1]}|, |\phi_{ik}^{[r]} - \phi_{ik}^{[r-1]}|, |\delta_{ik}^{[r]} - \delta_{ik}^{[r-1]}| \} < \varepsilon. \quad (45)$$

D. Computational complexity

Updating all variable-node messages requires $\mathcal{O}(N_\tau K)$ operations. The assignment-factor updates have the same complexity when the largest and second-largest ω_{ik} are retained for each bin.

Let M_k be the largest number of nondominated points retained in a Pareto frontier for vehicle k . Constructing the leave-one-out capacity frontiers independently gives the conservative time complexity $\mathcal{O}(\sum_{k \in \mathcal{K}} N_\tau^2 M_k)$. Prefix and suffix frontiers can be reused to reduce repeated computation in implementation.

Let n_k be the number of bins represented in the trellis of vehicle k , and let L_k be the largest number of feasible time, payload, and battery labels retained at one visited-set and location state. The forward and backward SOVA recursions require $\mathcal{O}(\sum_{k \in \mathcal{K}} L_k n_k^2 2^{n_k})$ time and $\mathcal{O}(\sum_{k \in \mathcal{K}} L_k n_k 2^{n_k})$ memory per cooperative round. The exponential term is confined to the bins represented by one vehicle trellis, and the K trellises can be processed in parallel. Traffic updates change $C_{uv}(t, \ell)$ and $D_{uv}(t, \ell)$ but do not alter the factor graph structure, allowing the retained messages and assignments to initialize the next replanning epoch.

IV. PERFORMANCE EVALUATION

This section evaluates the proposed CAP-trellis framework in a smart waste collection scenario constructed from a real urban road network. The evaluation addresses three questions: whether CAP enforces vehicle-capacity constraints under heterogeneous and time-varying demands, whether the trellis layer provides accurate and responsive regional routing, and whether both layers remain computationally practical as the service network grows.

TABLE I
SYSTEM PARAMETERS

Parameter	Value / Description
Number of smart bins, N	84
Baseline road network	From SUMO package
Number of vehicles, K	10
Curb mass, m_0	7,000 kg
Payload capacity, Q	2,000 kg
Maximum battery capacity, $B_{\max}$	180 kWh
Minimum battery reserve, $B_{\min}$	15 kWh
Rolling coefficient, C_{rr}	0.010
Frontal area of vehicle, A_f	5.5 m ²
Node weight, w_i	$\mathcal{U}(50, 250)$ kg
Propulsion efficiency, η_{drv}	0.90
Regenerative braking efficiency, η_{reg}	0.60
En-route auxiliary power, P_{aux}	3 kW
Service time per bin, T_s	120 s/bin
Stationary auxiliary power, P_{svc}	2 kW
Lifting/compaction energy per bin, e_i^{lc}	0.12 kWh/bin
Payload state interval, $\Delta \ell$	100 kg

A. Simulation setup

The evaluation uses a digital twin of the road network in Seongbuk-gu, Seoul, generated from OpenStreetMap data. The base instance contains $N = 84$ service nodes representing real waste-bin locations extracted from Seoul public data. Shortest-path road distances between all node pairs form the baseline cost matrix $\mathbf{D}_0$. As illustrated in Fig. 4(a), the digital twin preserves the street topology of the physical service area. Fig. 4(b) summarizes the three operating conditions considered in the simulations: uniform demand without disruption, random demand without disruption, and random demand with localized traffic disruptions. For the scalability experiments, nested instances are created by adding road-network nodes within the same boundary, increasing the problem size from 84 to 250 service nodes while retaining the original nodes.

Each service node i has demand w_i . Uniform weights are used when capacity represents the maximum number of nodes served by a vehicle, whereas independently sampled integer weights from $\{1, \dots, 9\}$ model heterogeneous waste volumes. In the disruption scenario, 5% of the nodes are

selected as disruption centers, and road costs within a 500-m radius are temporarily increased. Unless stated otherwise, results involving random demands or disruptions are averaged over 100 Monte-Carlo realizations.

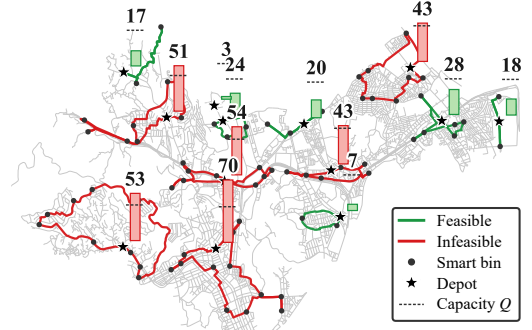
For district formation, CAP is compared with weighted affinity propagation (weighted AP) and weighted K -medoids using the same number of regions. Weighted AP scales the similarity as $w_{is}(i, j)$, and weighted K -medoids incorporates demand into medoid-based partitioning. These baselines account for demand in their objectives but do not explicitly enforce the regional capacity constraint. For routing, the proposed trellis is compared with brute force (BF), nearest neighbor (NN), and a genetic algorithm (GA). BF serves as the exact reference, NN is a greedy heuristic, and GA uses a population of 100, tournament selection, order crossover, swap mutation, and 500 generations. All route solvers operate on the same CAP partitions, isolating the effect of the routing algorithm.

B. Numerical results

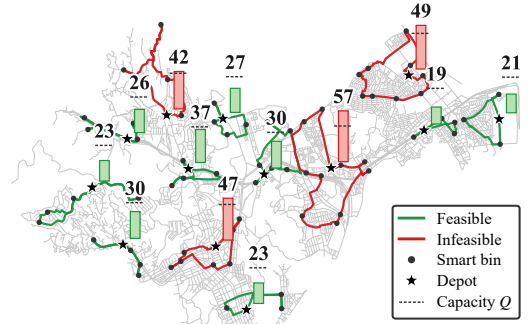
1) *Capacity feasibility under static demands*: The first experiment compares the feasibility of the regions generated by weighted affinity propagation, weighted K -medoids, and CAP under a common capacity budget $Q = 40$ and random node demands. A region is declared feasible when its aggregate demand satisfies $\sum_{i \in \mathcal{V}_k} w_i \leq Q$. As shown in Fig. 5, CAP produces 13 feasible regions out of 13, whereas weighted affinity propagation produces 9 feasible regions and weighted K -medoids produces 7. The maximum regional loads of the two baselines reach 57 and 70, respectively, demonstrating that demand-weighted similarities alone do not guarantee capacity feasibility.

This result reveals a fundamental difference between demand-aware clustering and capacity-constrained districting. The two weighted baselines tend to group geographically attractive nodes even when their combined demand exceeds the vehicle budget, because capacity is represented only indirectly through the clustering objective. CAP instead treats feasibility as an explicit condition during exemplar selection and assignment. Consequently, the improvement is not limited to a lower average load: it removes the overloaded tail of the regional-load distribution, which is the part that determines whether the resulting districts can actually be operated by the available vehicles.

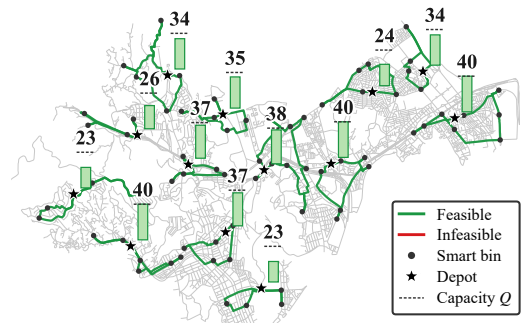
2) *Adaptation to time-varying service demands*: The second experiment evaluates districting under dynamically changing collection demands. The node weights evolve over 12 time steps, and 40% of the nodes experience a temporary demand surge by a factor between 2 and 4 during steps 5-7. Each method recomputes its partition at every step, and the served demand is classified according to whether it belongs to a capacity-feasible region. As shown in Fig. 6, CAP maintains zero infeasible demand over the complete horizon by increasing the number of active regions from 11 to 18 during the surge. In contrast, the cumulative infeasible demands of weighted affinity propagation and weighted K -medoids reach 1127 and 1684, respectively.



(a) Weighted K -medoids



(b) Weighted affinity propagation



(c) Capacitated affinity propagation

Fig. 5. Capacity-feasibility comparison under heterogeneous service demands.

The important observation is that CAP responds to the surge by changing the number and composition of districts rather than allowing the existing districts to become overloaded. This behavior makes the number of active regions an adaptive operational variable: additional vehicles or service rounds are introduced only while the demand level requires them. After the surge subsides, the district structure can contract again. The zero infeasible demand therefore indicates not only constraint satisfaction at isolated time steps but also the ability to track temporal demand variation without leaving a portion of the service demand uncovered.

3) *Regional routing performance*: The third experiment evaluates routing accuracy after fixing the district assignments obtained by CAP. As shown in Fig. 7, under uniform weights and $Q = 12$, BF and the proposed trellis produce the same total route length of 63.004 km, whereas GA and NN increase

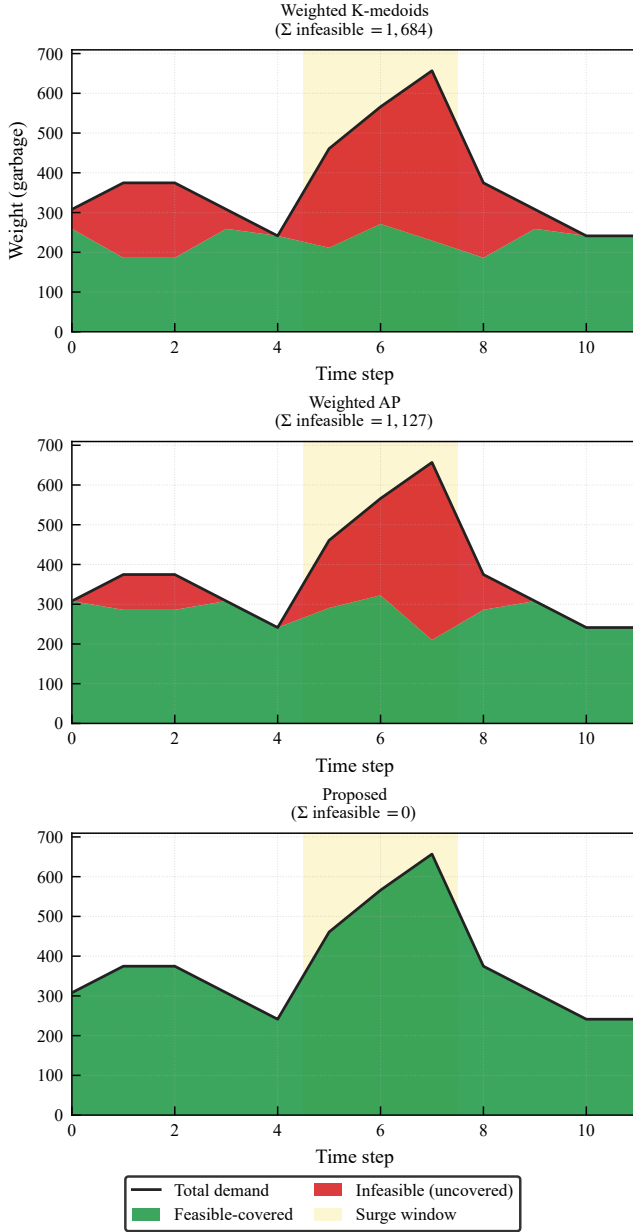
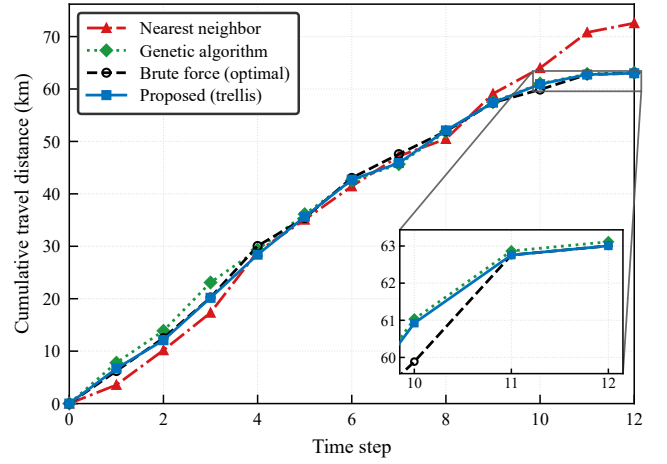


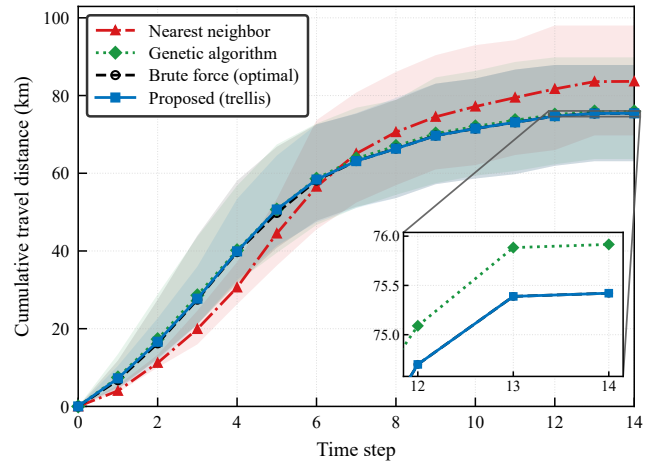
Fig. 6. Capacity-feasible and infeasible service demands under time-varying node weights.

the cost by 0.2% and 15.2%, respectively. Under random demands and $Q = 40$, the experiment is repeated over 100 Monte-Carlo realizations. The detailed comparison in Table 7c shows that the proposed trellis matches BF in every realization, yielding a mean total cost of 75.421 ± 5.585 km. GA produces a 0.7% higher cost, while NN produces a 10.9% higher cost. These results verify that the regional trellis computes the exact route for the evaluated cluster sizes.

Two insights follow from this comparison. First, the identical trellis and BF costs across all realizations confirm that the factor graph formulation preserves the original regional routing objective rather than approximating it. Second, the gap between NN and the exact methods remains substantial under both uniform and random demands, showing that geographically plausible greedy decisions can create expensive



(a) Uniform demands



(b) Random demands

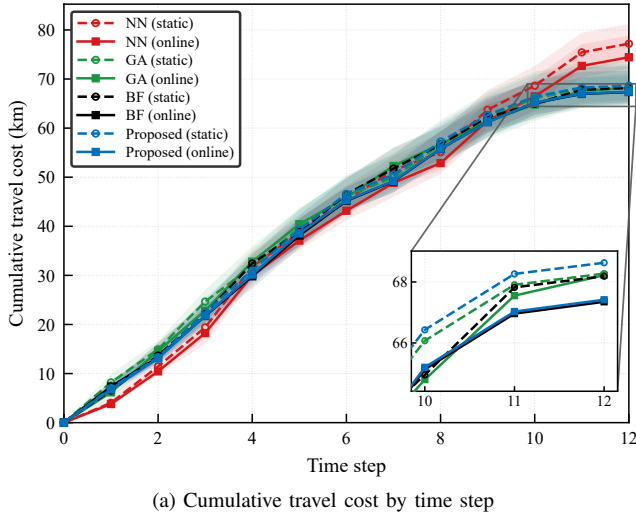
Method	Mean total cost	vs Proposed (%)
NN	83.642 (± 6.108)	+10.9
GA	75.916 (± 5.674)	+0.7
BF	75.421 (± 5.585)	+0.0
Proposed	75.421 (± 5.585)	-

(c) Performance table

Fig. 7. Cumulative regional routing cost under uniform and random service demands.

downstream connections. GA approaches the optimum more closely, but its residual gap and iterative search cost remain unnecessary for the capacity-bounded regional problems considered here. The trellis therefore provides exactness without relying on repeated stochastic search.

4) *Online adaptation to road disruptions*: The fourth experiment evaluates online route adaptation under localized disruptions. At every time step, 5% of the service nodes are selected as disruption epicenters. The costs of road edges within 500 m of each epicenter are multiplied by 3, and each disruption remains active for three consecutive time steps. Static routing commits to the route computed before departure, whereas online routing recomputes the remaining open path from the current vehicle location using the active cost matrix. The results are averaged over 100 disruption realizations.



Method	Mean total cost (km)		Saving (%)	Replanning time (s)
	Static ($\pm$ std)	Online ($\pm$ std)		
NN	77.196 (± 1.551)	74.440 (± 1.211)	3.6	~ 0
GA	68.273 (± 1.601)	68.220 (± 1.255)	0.1	~ 93
BF	68.189 (± 1.537)	67.351 (± 1.297)	1.2	4.5
Proposed	68.630 (± 1.641)	67.417 (± 1.309)	1.8	0.13

(b) Performance table

Fig. 8. Static and online routing performance under localized road disruptions.

Figure 8 compares the cumulative travel costs of static and online routing under localized disruptions. Online re-optimization reduces the mean travel cost of the proposed method from 68.630 to 67.417 km, corresponding to a 1.8% saving. As summarized in Table 8b, the online BF solution attains a slightly lower cost of 67.351 km, while requiring substantially greater computation. NN achieves the largest relative saving of 3.6%, but its online cost remains 74.440 km and is therefore considerably higher than those of the exact and trellis-based methods. GA provides only a 0.1% improvement because its population is reinitialized at every routing step. The proposed method therefore preserves near-BF routing quality while supporting rapid online replanning.

The relative saving alone does not fully describe online routing quality. NN exhibits the largest percentage reduction because its static route has the highest initial cost, yet its replanned route remains worse than those of BF and the trellis. A more relevant measure is therefore the combination of final travel cost and replanning latency. The proposed method finishes within 0.066 km of online BF while reducing the replanning time from 4.5 s to 0.13 s. This quality-latency balance is particularly important when disruptions change faster than a computationally expensive solver can repeatedly produce new routes.

5) *Scalability of district formation*: The fifth experiment increases the number of service nodes from 84 to 250 while keeping the demand associated with each existing node fixed. The regional routes are computed using the exact trellis, and only the districting method is varied. The performance metric is the fraction of total demand assigned to capacity-feasible regions. The districting results are presented in Fig. 9a,

which corresponds to the upper panel of the overall scalability comparison in Fig. 9. CAP serves 100% of the demand over the complete range by increasing the number of active regions as the service area grows. In contrast, the feasible-demand ratio of weighted affinity propagation decreases from 56.6% to 11.6%, while that of weighted K -medoids decreases from 41.3% to 32.5%. The widening gap shows that capacity violations accumulate as the service network grows when the constraint is not embedded directly into the districting procedure.

The scalability trend indicates that feasibility errors in the uncapacitated baselines compound as new nodes are introduced. Although their clustering objectives remain demand-aware, they do not provide a mechanism that forces additional regions to appear when the total service load grows. CAP, by contrast, converts network growth into an increase in the number of capacity-bounded districts, keeping the feasible-demand ratio unchanged. This distinction is operationally significant because expanding the service area should require more vehicle assignments, not progressively larger violations within a fixed or poorly adjusted district structure.

6) *Scalability of regional routing*: The sixth experiment fixes the CAP partitions and compares the regional route solvers over the same nested instances. As shown in Fig. 9b, the proposed trellis and BF return identical route costs for all 27 evaluated instances. GA remains within 0.03%-0.31% of the optimum, whereas NN incurs an excess cost of 7.9%-11.4%. The computation-time results in Fig. 9b show that the trellis time increases from 0.09 s at $N = 84$ to 0.30 s at $N = 250$, reflecting the approximately linear increase in the number of capacity-bounded regions. Over the same range, the BF time increases from 0.36 s to 13.67 s and the GA time from 19.5 s to 54.2 s. The trellis is faster than BF in 26 of the 27 instances, with a median speedup of 5.6 and a maximum speedup of 63.7.

This scaling behavior results from decomposing the global service network into capacity-bounded regional problems. As N increases, CAP primarily increases the number of districts rather than allowing the size of each routing problem to grow without limit. The trellis can therefore solve more small exact subproblems with a modest increase in total time, whereas BF remains sensitive to the combinatorial size of individual tours. The result explains why the proposed framework can preserve exact regional optimality while exhibiting substantially better runtime growth than exhaustive search over the tested network sizes.

Overall, the results show that embedding the capacity constraint directly into the districting messages is necessary to preserve feasibility under heterogeneous, time-varying, and large-scale service demands. The exact regional trellis matches brute-force routing over all tested instances while providing substantially lower computation time as the regional size increases. The coupled framework therefore combines capacity-feasible district formation, exact regional routing, and practical online adaptation within a common distributed architecture.

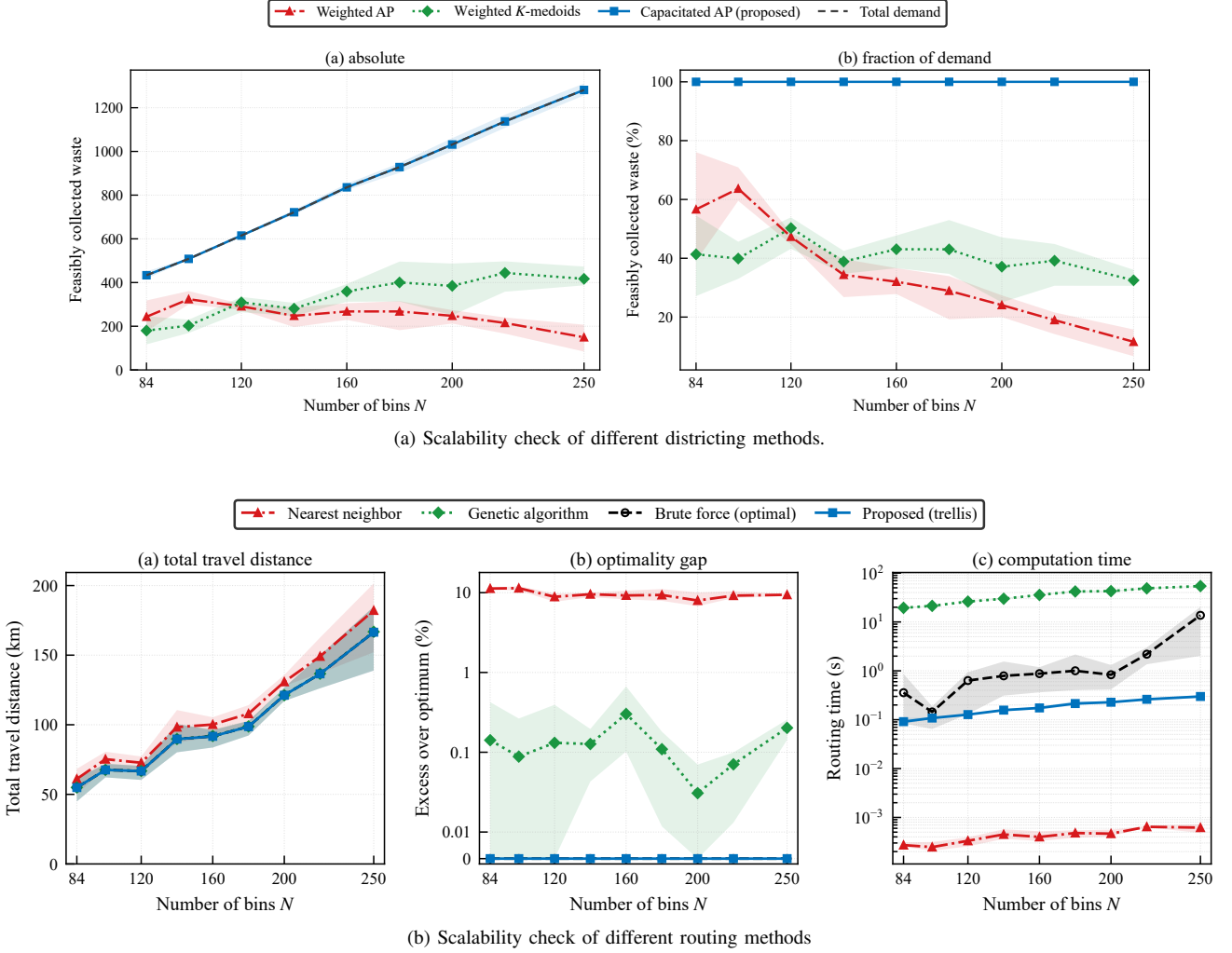


Fig. 9. Scalability comparison.

V. CONCLUSION

This paper developed a distributed message-passing framework that jointly determines capacity-feasible service districts and regional routes for smart waste collection. By coupling CAP with independent routing trellises through sparse bridging factors, the framework allows route-level information to influence district formation while retaining parallel and scalable computation. Simulations on a real urban road network showed that the proposed method preserves capacity feasibility and exact regional routing performance over the evaluated instances, while enabling substantially faster online replanning under changing operating conditions. These results demonstrate the potential of the proposed architecture as a practical basis for adaptive urban fleet coordination.

APPENDIX A PARETO-FRONTIER EVALUATION

The capacity factor V_k receives γ_{jk} from every bin variable connected to vehicle k . After excluding b_{ik} , the unrestricted

capacity subproblem is

$$P_k^{-i}(L) = \max_{\substack{b_{jk} \in \{0,1\} \\ j \in \mathcal{U}_\tau \setminus \{i\}}} \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} \gamma_{jk} b_{jk} \\ \text{subject to } \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} w_j b_{jk} \leq L. \quad (46)$$

The corresponding nonempty subproblem is

$$P_{k,+}^{-i}(L) = \max_{\substack{b_{jk} \in \{0,1\} \\ j \in \mathcal{U}_\tau \setminus \{i\}}} \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} \gamma_{jk} b_{jk} \\ \text{subject to } \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} w_j b_{jk} \leq L, \\ \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} b_{jk} \geq 1. \quad (47)$$

Both are 0/1 knapsack problems. Since (27) queries $P_k^{-i}(q_k - w_i)$ and $P_{k,+}^{-i}(q_k)$ using the same incoming messages, their Pareto frontiers are constructed in parallel and reused.

The unrestricted frontier is initialized with the point $(0, 0)$. For every bin $j \neq i$, the current frontier is merged with a shifted copy obtained by adding w_j to each retained weight and γ_{jk} to its associated value. Points exceeding q_k are

removed, equal-weight points retain only the largest value, and dominated points are discarded. After all bins have been processed, $P_k^{-i}(L)$ is obtained from the retained point with the largest value among those whose weight does not exceed L .

For $P_{k,+}^{-i}(L)$, a separate nonempty frontier is maintained in parallel and initialized as empty. For every bin $j \neq i$, the current nonempty frontier is merged with its shifted copy and the singleton point (w_j, γ_{jk}) . The same capacity filtering, equal-weight reduction, and dominance pruning are applied. After all bins have been processed, $P_{k,+}^{-i}(L)$ is obtained from the retained point with the largest value among those whose weight does not exceed L . If no such point exists, then $P_{k,+}^{-i}(L) = -\infty$.

These are exact Nemhauser–Ullmann procedures [28] applied to the unrestricted and nonempty selections, respectively.

APPENDIX B

DERIVATION OF MESSAGE UPDATE RULES

For $b_{ik} = 1$, factor I_i forces every b_{ih} with $h \neq k$ to zero. For $b_{ik} = 0$, exactly one of the remaining variables must equal one. Thus,

$$\mu_{I_i \rightarrow b_{ik}}(1) = \sum_{h \in \mathcal{K} \setminus \{k\}} \mu_{b_{ih} \rightarrow I_i}(0) \quad (48)$$

$$\mu_{I_i \rightarrow b_{ik}}(0) = \sum_{h \in \mathcal{K} \setminus \{k\}} \mu_{b_{ih} \rightarrow I_i}(0) + \max_{h \in \mathcal{K} \setminus \{k\}} \omega_{ih}. \quad (49)$$

Subtracting the second message from the first gives

$$\eta_{ik} = - \max_{h \in \mathcal{K} \setminus \{k\}} \omega_{ih}. \quad (50)$$

For $b_{ik} = 1$, the remaining capacity is $q_k - w_i$. For $b_{ik} = 0$, the remaining bins may use the full capacity q_k . After the state-zero terms common to both messages are removed,

$$\mu_{V_k \rightarrow b_{ik}}(1) = P_k^{-i}(q_k - w_i) \quad (51)$$

$$\mu_{V_k \rightarrow b_{ik}}(0) = P_k^{-i}(q_k). \quad (52)$$

Their difference yields

$$\phi_{ik} = P_k^{-i}(q_k - w_i) - P_k^{-i}(q_k). \quad (53)$$

APPENDIX C

SOFT-OUTPUT DYNAMIC PROGRAMMING ALGORITHM

At replanning epoch τ , the vehicle- k trellis is initialized from the observed vehicle state $\mathbf{y}_k(t_\tau)$. Since no remaining bin has yet been visited in the candidate route, the initial state message is

$$\alpha_0^{(k)}(\mathbf{x}_0^{(k)}) = \begin{cases} 0, & \mathbf{x}_0^{(k)} = [\mathbf{0}^\top \quad t_\tau \quad \mathbf{y}_k(t_\tau)^\top]^\top \\ -\infty, & \text{otherwise.} \end{cases} \quad (54)$$

Applying the max-sum factor update at $G_h^{(k)}$ gives

$$\psi_h^{(k)}(\mathbf{x}_h^{(k)}) = \max_{\mathbf{x}_{h-1}^{(k)}} [G_h^{(k)}(\mathbf{x}_{h-1}^{(k)}, \mathbf{x}_h^{(k)}) + \alpha_{h-1}^{(k)}(\mathbf{x}_{h-1}^{(k)})]. \quad (55)$$

Since an internal state node has only two neighboring transition factors, its forward output is

$$\alpha_h^{(k)}(\mathbf{x}_h^{(k)}) = \psi_h^{(k)}(\mathbf{x}_h^{(k)}), \quad h = 1, \dots, H_{k,\tau} - 1. \quad (56)$$

The terminal state similarly gives

$$\zeta_k(\mathbf{x}_{H_{k,\tau}}^{(k)}) = \psi_{H_{k,\tau}}^{(k)}(\mathbf{x}_{H_{k,\tau}}^{(k)}). \quad (57)$$

At R_k , the message to the terminal state maximizes over all adjacent assignment variables,

$$\lambda_k(\mathbf{x}_{H_{k,\tau}}^{(k)}) = \max_{\mathbf{b}_{:,k}} \left[R_k(\mathbf{b}_{:,k}, \mathbf{x}_{H_{k,\tau}}^{(k)}) + \sum_{i \in \mathcal{U}_\tau} \rho_{ik} b_{ik} \right]. \quad (58)$$

This message becomes the terminal input of the backward recursion,

$$\xi_{H_{k,\tau}}^{(k)}(\mathbf{x}_{H_{k,\tau}}^{(k)}) = \lambda_k(\mathbf{x}_{H_{k,\tau}}^{(k)}). \quad (59)$$

Applying the max-sum factor update in the reverse direction gives

$$\beta_{h-1}^{(k)}(\mathbf{x}_{h-1}^{(k)}) = \max_{\mathbf{x}_h^{(k)}} [G_h^{(k)}(\mathbf{x}_{h-1}^{(k)}, \mathbf{x}_h^{(k)}) + \xi_h^{(k)}(\mathbf{x}_h^{(k)})], \quad (60)$$

and the internal state update is

$$\xi_h^{(k)}(\mathbf{x}_h^{(k)}) = \beta_h^{(k)}(\mathbf{x}_h^{(k)}), \quad h = 1, \dots, H_{k,\tau} - 1. \quad (61)$$

Finally, fixing $b_{ik} = b$ in the routing factor gives

$$\begin{aligned} \mu_{R_k \rightarrow b_{ik}}(b) = & \max_{\mathbf{b}_{:,k} \setminus b_{ik}, \mathbf{x}_{H_{k,\tau}}^{(k)}} \left[R_k(\mathbf{b}_{:,k}, \mathbf{x}_{H_{k,\tau}}^{(k)}) + \zeta_k(\mathbf{x}_{H_{k,\tau}}^{(k)}) \right. \\ & \left. + \sum_{j \in \mathcal{U}_\tau \setminus \{i\}} \rho_{jk} b_{jk} \right]. \end{aligned} \quad (62)$$

The difference between the two conditioned maxima is

$$\delta_{ik} = \mu_{R_k \rightarrow b_{ik}}(1) - \mu_{R_k \rightarrow b_{ik}}(0). \quad (63)$$

Because $G_h^{(k)}$ accumulates negative energy, this difference measures the energy-aware preference for including bin i in the route of vehicle k after accounting for the incoming assignment messages.

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